Sharper forecasts, longer promises: calibration transfer in delivery-date policies

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1. Summary

A sharper forecasting model did not give shorter promised delivery dates, because the calibration learned in one period carried over differently to the next.

I started this project wanting to predict which Olist orders would arrive late. Along the way I realised that "late" compares two things: how long delivery takes, and how generous the promised date is. The question became: **when does a better forecast actually give a better promised date?** The data are public Olist e-commerce orders (Brazil, 2017–2018), restricted to single-seller orders that were eventually delivered. A promise is a whole number of days; an order is covered if it arrives in time; the target coverage is 90%.

I compared a per-state quantile rule, a linear quantile regression and a shallow gradient-boosted model (HGB) under a protocol written down before any new model was fitted. Three findings:

1. **The comparison design changed the verdict.** In the project's earlier comparison, regression promised 2.57 more days than the state rule. With the same training data, the same calibration procedure and the same feature timing, it promised 1.54 fewer days at 0.31 points lower coverage.
2. **HGB is the sharper model, yet its calibrated promise was longer.** At equal coverage HGB needs fewer days than the linear model: 0.14 days fewer on the calibration period and 0.6–0.9 days fewer on the evaluation period (post hoc). But the shift learned on the calibration period carried HGB to 97.1% coverage on the evaluation period, against 96.2% for the linear model. Its registered promise was therefore 1.22 days longer.
3. **No policy met the pre-set bar for replacing the existing rule.** This verdict rests on the calibration level. The rule gives no credit for extra coverage, and at the existing rule's own coverage both models need fewer days (post hoc).

The evaluation orders had already been examined earlier in the project, so this is a pre-registered exploratory study, not a blind test. Every analysis added after the scoring run is labelled post hoc and changes no registered verdict.

2. Loss and decision: why a quantile, not the mean

If a late day and a spare day cost different amounts, the best promise is a quantile of the delivery time, not its mean. Let Y be the delivery time in whole days and d the promise, with the loss

$$L(d, Y) = c_{late} \cdot (Y - d)_+ + c_{slack} \cdot (d - Y)_+.$$

Let R(d) = E[L(d, Y)] and F the distribution function of Y. Moving the promise from d to d + 1 saves c_late on every order with Y > d and adds c_slack on every order with Y ⩽ d. Therefore

$$R(d + 1) - R(d) = (c_{late} + c_{slack}) \cdot F(d) - c_{late}.$$

This difference never decreases in d, so R is minimised at the smallest whole d with F(d) ⩾ q, where q = c_late / (c_late + c_slack). That is the empirical quantile rule the project uses (NumPy's inverted_cdf). The pinball loss at level q is this loss with c_late = q and c_slack = 1 − q (scikit-learn); a 90% target thus corresponds to a 9 : 1 penalty ratio. The mean minimises squared error and the median a 1 : 1 absolute loss; both treat a late day and a spare day alike.

A check on validation data. On the 12,045 validation orders I gave every order the same promise and computed the 9 : 1 loss for d = 1, …, 60. The minimum is at d = 20, the empirical 90% quantile: F(19) = 89.6% < 90% ⩽ F(20) = 91.4%. Promising the mean (11.43, rounded up to 12) would have covered only 65.5% of orders, at a loss of 23.76 per order against 15.61.

What 9 : 1 does not mean. It follows from choosing q = 0.90; I did not measure it, and it is not Olist's real cost ratio. Customer harm may not grow linearly with late days; if one late day already does most of the damage, a coverage constraint is closer to the truth than a linear cost. Therefore the study reports the trade-off between coverage and promised days. It does not claim a profit-optimal date.

3. Comparison design

A fair comparison changes only the method. In the fair comparison, each calibrated policy used the same training orders, the same label cut-off, the same rule for seller-history features and the same calibration procedure; the uncalibrated controls (B1 and the retained rule B0) are kept apart. A label counts only if the order was delivered before the decision date.

Segment	Purchases	Orders	Late rate*	Role
F0 → V	Jan–Jun → Jul–Sep 2017	13,616 / 12,045	3.89% / 3.38%	fit and select settings
F	Jan–Sep 2017	25,666	3.67%	final fit
C	Oct–Dec 2017	17,056	8.74%	calibration shift
E	Apr–Jul 2018	19,950	4.26%	final scoring run

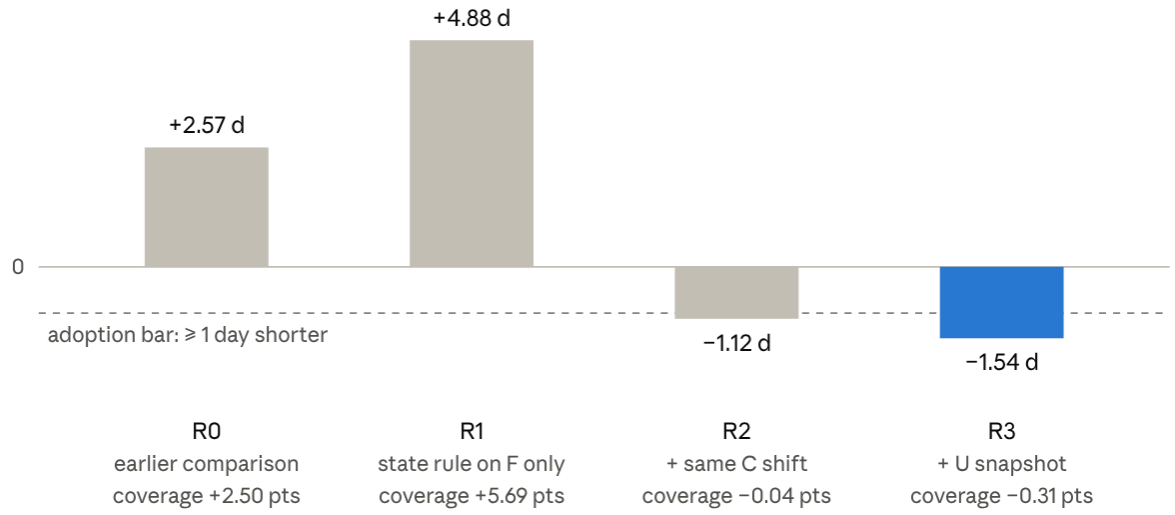
*Against Olist's own estimated date.

Calibration. Each calibrated policy is shifted by its own number computed on C: the k-th smallest of its residuals $y - \hat{y}$, with $k = \lceil (n + 1) \cdot 0.9 \rceil$ and $n = 17{,}056$. The promise is then rounded up: $\max\{1, \lceil \hat{y} + \text{shift} \rceil\}$. This has the form of split conformal, but its guarantee needs C and E to be exchangeable, and C had a much higher late rate than E (Barber et al.). Section 5 explains why rounding after the shift gives continuous models extra slack.

Stepwise comparison. The earlier comparison (R0) differed in three conditions at once, so I changed one condition per step. The pre-set primary comparison is R3 against R0; R1 and R2 are descriptive.

Making the comparison fair flipped the gap from +2.57 to −1.54 days

Regression minus state rule, mean promised days on E (19,950 orders); one condition changed per step

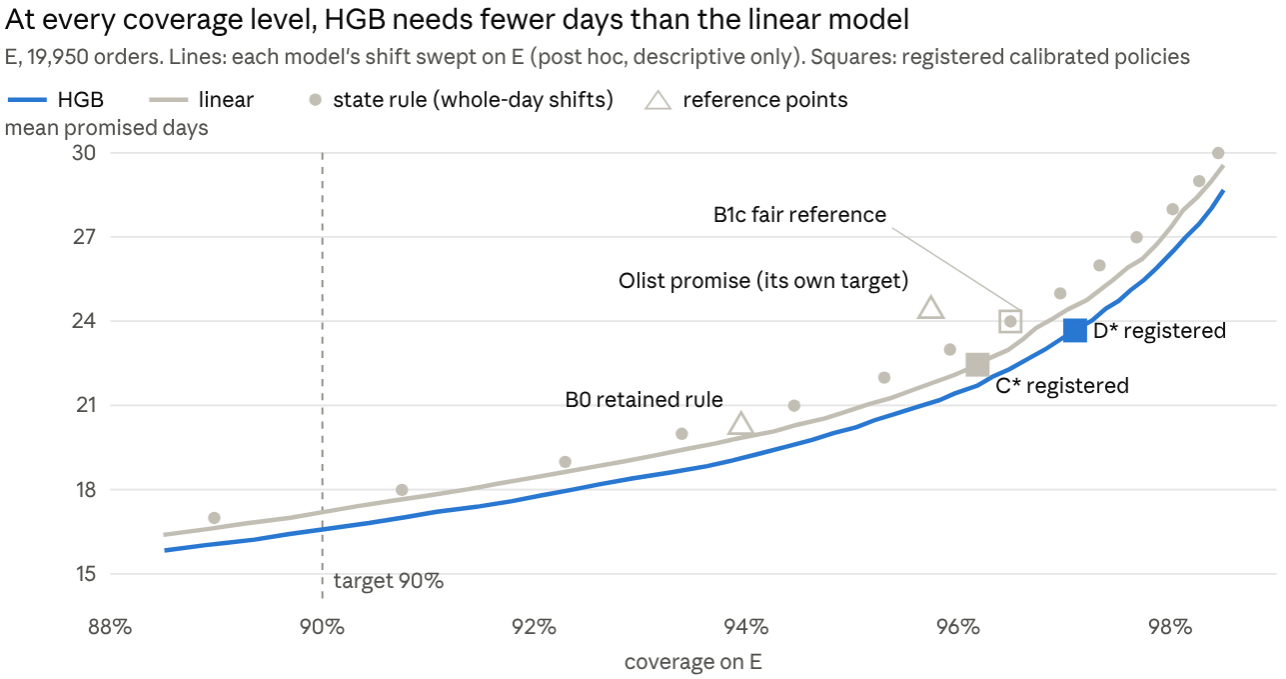


outputs/e1/score/h1_steps.csv · E, April–July 2018, 19,950 orders

Under this fixed order, changing the design alone reversed the judgement; the largest step is giving the state rule the same calibration shift (+6 days). This is design sensitivity, not a causal statement about which method is better. Two references are kept apart: the fair reference B1c (state rule on F with the same calibration) and the replacement reference B0 (the project's retained rule, fitted on F + C without a shift).

4. Results

The E1 models and decision rules were frozen before the final scoring run on E, and each policy was scored once in that run. The figure shows the trade-off that the registered verdicts compress into single points. Each line sweeps one model's shift over E to show how many days it needs for each coverage level. This sweep uses E itself, so it only compares models at equal coverage; it selects nothing.



outputs/e1/posthoc_r3/days_coverage_curves_E.csv and outputs/e1/score/policy_table.csv · E, 19,950 orders · curves are post hoc and choose no policy

At 90% coverage HGB needs 16.60 days and the linear model 17.20; at 96.18% (where the registered linear policy landed) they need 21.71 and 22.44. HGB is the sharper model. Its registered policy sits higher only because the shift learned on C carried it further right, to 97.1% coverage. The same holds on C itself: at the registered shifts, HGB covered 91.14% with 24.54 days and the linear model 91.19% with 24.68 days.

The registered verdicts are unchanged. Intervals are week-block bootstrap intervals over 17 purchase weeks of unequal size, so they are approximate.

Question	Comparison	Registered result	Verdict
H1: does the fair design change the judgement?	R3 vs R0	gap moved by −4.11 days	met
H2: does a linear change help?	C* vs B3	no candidate passed selection on V, so C* = B3	not tested (registered as "not supported")

Question	Comparison	Registered result	Verdict
H3: does shallow HGB help?	D* vs B1c	−0.32 days (about −0.55 to −0.17), coverage +0.61 pts	below the 1-day bar
Replace the retained rule?	C*, D* vs B0	+2.15 and +3.37 days	no

At q = 0.9126, a target matching Olist's coverage on C, the linear policy promised 23.42 days at 96.62% coverage against Olist's 24.44 days at 95.74%. It passes the pre-set rule by 0.02 days, and a post-hoc interval for the day difference (−3.25 to +0.69) includes zero. Section 5 adds a further reason for caution: the evaluation sample is filtered on Olist's own estimated date.

5. What explains the gap

Four features of the design move the registered points. All numbers below are post hoc: they reuse the saved predictions, choose no new policy and change no registered verdict.

Sensitivity	What it is	Effect	Note
Rounding after the shift	The shift is the k-th smallest residual $y - \hat{y}$, and the promise is then rounded up. For a continuous $\hat{y}$ this adds slack that the integer state rule does not get.	On C, the continuous models cover 91.1–91.2% instead of 90%; the state rule covers 90.3%. An integer-aware shift (k-th smallest of $y - 1 - \hat{y}$) makes every continuous promise exactly 1 day shorter. On E, HGB would then be 1.32 days shorter than B1c at +0.18 points coverage.	a flaw in the registered calibration formula; the registered H3 verdict stands
Calibration season	C runs from October to December 2017 and includes the November peak.	Late rates on C were 4.2%, 12.5% and 7.4% by month; November holds 42% of C. A shift learned there is likely to over-cover a calmer period.	a candidate cause of over-coverage, not tested separately

Sensitivity	What it is	Effect	Note
Evaluation window	The project's population keeps orders whose Olist estimated delivery week is no later than 2018-07-23. Late-June and July orders therefore enter only if Olist promised a short time.	July has 1,085 orders, Olist promises of at most 26 days (56–63 in April–June) and a mean actual time of 6.2 days. April–May only: linear 95.1%, HGB 96.4%, B1c 95.5% coverage; June–July 98.3–98.5%.	part of the over-coverage; the Olist comparison is filtered on the comparator itself
Replacement rule	A policy may lose up to 1 point of coverage but gets no credit for extra coverage.	At B0's own coverage (93.95%) on E, HGB needs 19.12 days and the linear model 19.83, against 20.31 for B0.	the "do not replace" verdict reflects the calibration level more than the models

These four points do not overturn the registered results. They change how the results should be read: the models differed less than the calibration did.

6. Limits and conclusion

Limits. One evaluation period, already examined earlier in the project; a conditional population (single-seller, eventually delivered orders); a small stepwise model search; approximate intervals from 17 week clusters. Promised dates do not change logistics, so nothing here shows faster delivery or happier customers.

Conclusion. The pre-registered verdicts stand: the fair design reversed the regression-versus-rule judgement, no linear change was selected, HGB did not clear the 1-day bar, and nothing replaced the existing rule. Read together with the sensitivity checks, the lesson is narrower and more useful than "complex models do not help". HGB is the sharper model, but a calibration shift learned in a peak season, rounded in a way that favours integer rules, and evaluated on a window filtered by Olist's own dates, decided where each policy landed. The next question is therefore about calibration, not about models: an integer-aware shift and recalibration that tracks changing conditions, tested on a new, separately registered evaluation.

Supplementary material (model selection, the simulation, all post-hoc sensitivity tables), the integrity record and a contribution statement: Supplement (a separate PDF by email).